



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<Name>

<Date>



Outline

- **Executive Summary**
- **Introduction**
- **Methodology**
- **Results**
- **Conclusion**
- **Appendix**

Executive Summary

Summary of Methodologies:

Various Machine Learning models were used to classify, based on a selection of features, a successful landing. These include Logistic Regression, Support Vector Machines, Decision Tree and KNN. These were all Cross Validated and tuned to optimal hyperparameters.

Summary of Results:

All the ML models produced similar results given the limited data set. All achieving accuracy scores of 83%+, giving a robust evaluation of the classification models.

Introduction

Context:

SpaceX has gained worldwide attention for a series of historic milestones.

It is the only private company ever to return a spacecraft from low-earth orbit, which it first accomplished in December 2010. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars whereas other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage.

Goals:

Therefore if we can determine if the first stage will land, we can determine if SpaceX will reuse the first stage, and therefore determine the cost of the launch.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - We will be working with SpaceX launch data that is gathered from an API, specifically the SpaceX REST API. This API will give us data about launches, including information about the rocket used, payload delivered, launch specifications, landing specifications, and landing outcome. We will be specifically looking at Falcon 9 Booster versions, and we will webscrape historic launch data from Wikipedia to supplement the API data
- Perform data wrangling
 - The column Outcome indicates if the first stage successfully landed. There are 8 outcomes to be converted to Classes to Binary (either 0 or 1). 0 is a bad outcome, that is, the booster did not land. 1 is a good outcome, that is, the booster did land. The variable Y will represent the classification variable that represents the outcome of each launch.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Models used include Logistic Regression, Support Vector Machines, Decision Tree and KNN. These were all Cross Validated and tuned to optimal hyperparameters.

Data Collection

- We will be working with SpaceX launch data that is gathered from an API, specifically the SpaceX REST API. This API will give us data about launches, including information about the rocket used, payload delivered, launch specifications, landing specifications, and landing outcome.
- We will be specifically looking at Falcon 9 Booster versions, and we will webscrape historic launch data from Wikipedia to supplement the API data using Python Get responses function

Data Collection – SpaceX API

Fetch Data:

- Send GET request to SpaceX API URL.
- Decode JSON and convert to Pandas DataFrame.

Prepare Data:

- Select relevant columns (rocket, payloads, launchpad, cores, flight_number, date_utc).
- Filter rows with multiple cores or payloads, convert date format, and filter by launch date ($\leq 2020-11-13$).

Core Data: Extract landing outcome, reuse, and more.

Create Final Dataset:

- Combine extracted data into a new DataFrame.
- Filter for Falcon 9 launches, reset FlightNumber.

Handle Missing Data:

- Replace missing PayloadMass with its mean.
- Keep None for LandingPad.

Export Data:

- Save to CSV (dataset_part_1.csv)

API Call Workflow Summary:

Start

|

v

Send GET Request → Check Status (200) →
Decode JSON → Convert to DataFrame

|

v

Filter Columns & Rows → Convert Date →
Extract Data (Booster, Payload, Launch Site,
Core)

|

v

Filter for 'Falcon 9' → Handle Missing
Values → Save to CSV

Notebook address:

https://github.com/ChiLau-jm/IBM_coursera/blob/ChiLau-jm-patch-1/Space%20X%20Capstone%20pt1b%20-%20API%20calls.ipynb

Data Collection - Scraping

Request Wiki Page:

- Fetch Falcon9 Launch page with requests.get() and parse with BeautifulSoup.

Extract Columns:

- Find all tables and extract column names from the third table header.

Create Dictionary:

- Initialize launch_dict with columns and empty lists, then remove irrelevant ones.

Parse Launch Data:

- Extract flight details (date, time, payload, etc.) from table rows.

Fill Dictionary:

- Append parsed values to launch_dict.

Create & Export DataFrame:

- Convert launch_dict to DataFrame and export to CSV (spacex_web_scraped.csv).

Webscrape Workflow Summary:

Request & Parse Wiki Page

|

v

Extract Column Names from 3rd table header

|

v

Create Dictionary with columns and empty lists → Parse Data from rows and fill dictionary

|

v

Create DataFrame and Export to CSV

Data Wrangling

Calculate Orbit Frequencies:

- Use `.value_counts()` to determine the frequency of each orbit type in the Orbit column of the DataFrame.

Count Mission Outcomes:

- Apply `.value_counts()` on the Outcome column to calculate the number and occurrence of each mission outcome (e.g., successful or failed landings).

Identify Bad Outcomes:

- Extract and create a list of bad landing outcomes based on specific indices (e.g., unsuccessful ocean, RTLS, or ASDS landings).

Create Landing Classification:

- Generate a new column `landing_class` where the value is 0 for bad outcomes and 1 for successful landings. This is used to classify whether the mission's first stage landing was successful.

Calculate Success Rate:

- Calculate the mean of the `landing_class` column to determine the success rate of the launches, and export the DataFrame to a CSV file for further analysis.

EDA with Data Visualization

Chart 1: Flight Number vs Launch Site (Scatter Plot)

Suitable for identifying trends or clusters in flight numbers across different launch sites. A scatter plot helps visualize the relationship between a numeric variable (flight number) and a categorical variable (launch site).

Chart 2: Payload vs Launch Site (Scatter Plot)

Useful for spotting patterns in payload sizes at each launch site. The scatter plot effectively shows how payloads vary by launch site, highlighting outliers or trends.

Chart 3: Success Rate of Each Orbit Type (Bar Chart)

A bar chart is ideal for comparing the success rates of different orbit types. It makes it easy to compare categorical data (orbit types) and their associated success rates.

Chart 4: Flight Number vs Orbit Type (Scatter Plot)

The scatter plot helps visualize how flight numbers are distributed across different orbit types, revealing any trends or frequency differences.

Chart 5: Payload vs Orbit Type (Scatter Plot)

A scatter plot is effective in showing the relationship between the payload and orbit type, identifying if certain orbits carry heavier or lighter payloads.

Chart 6: Launch Success Yearly Trend (Line Chart)

A line chart is perfect for showing trends over time, such as the yearly launch success rate. It helps track improvements or declines in success rate across years.

EDA with SQL

- **Unique Launch Sites:**
 - Use `SELECT DISTINCT "Launch_Site" FROM "SPACEXTABLE"` to display all unique launch sites in the "SPACEXTABLE".
- **Filter Launch Sites Starting with 'CCA':**
 - Use `SELECT * FROM "SPACEXTABLE" WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5` to list 5 records where the launch site starts with 'CCA'.
- **Payload Mass for NASA (CRS):**
 - Use `SELECT SUM("PAYLOAD_MASS_KG_") FROM "SPACEXTABLE" WHERE "Customer" LIKE 'NASA (CRS)'` to calculate the total payload mass carried by NASA (CRS).
- **Average Payload Mass for 'F9 v1.1' Booster Version:**
 - Use `SELECT AVG("PAYLOAD_MASS_KG_") FROM "SPACEXTABLE" WHERE "Booster_Version" LIKE 'F9 v1.1%'` to find the average payload mass for the 'F9 v1.1' booster.
- **First Successful Ground Pad Landing:**
 - Use `SELECT * FROM "SPACEXTABLE" WHERE "Landing_Outcome" LIKE 'Success (ground pad)' ORDER BY "Date" ASC LIMIT 1` to list the date of the first successful ground pad landing.
- **Failed Drone Ship Landings in 2015:**
 - Use `SELECT *, TO_CHAR("Date"::timestamp, 'Month') AS "Month_Name" FROM "SPACEXTABLE" WHERE TO_CHAR("Date"::timestamp, 'YYYY') = '2015' AND "Landing_Outcome" IN ('Failure (drone ship)')` to display records of failed drone ship landings in 2015, including the month name.

Build an Interactive Map with Folium

- **Download and Visualize Launch Sites:**

- Load the `spacex_launch_geo.csv`, create a Folium map, and mark each launch site's coordinates using `folium.Circle` and `folium.Marker`.

- **Mark Launch Outcomes:**

- Add success/failure markers to the map with color differentiation (green for success, red for failure), and use `MarkerCluster` to group overlapping markers.

- **Calculate Distances:**

- Use `MousePosition` in Folium to get coordinates and calculate distances from launch sites to coastlines using the Haversine formula.

- **Visualize Coastline Proximity:**

- Add a marker for the closest coastline and display the calculated distance between the launch site and coastline.

- **Draw Polyline:**

- Create a `folium.PolyLine` to visually connect the launch site to the coastline.

- **Analyze Launch Site Distribution:**

- Explore the geographic distribution of launch sites, especially their proximity to the equator and coastlines for insights.

Build a Dashboard with Plotly Dash

The Dashboard contains the following components:

- **Dropdown for Launch Site Selection:** A dcc.Dropdown component allows users to select a specific launch site or choose "All Sites" to view data from all locations. This interaction drives the update of the plots.
- **Pie Chart** A dcc.Graph is used to display a pie chart showing the count of successful launches. The chart dynamically updates based on the selected launch site, either showing data for all sites or filtering it for a specific one. The pie chart is suitable because it clearly shows proportions, which is ideal for visualizing success counts.
- **RangeSlider for Payload Mass:** The dcc.RangeSlider allows users to select a range for payload mass, enabling them to filter the data based on the payload's weight. This slider enhances interactivity by enabling users to focus on specific payload ranges.
- **Scatter Plot:** Another dcc.Graph displays a scatter plot that correlates payload mass with launch success. It colors the data points by "Booster Version Category" to distinguish between different boosters. The scatter plot is suitable for showing trends and relationships between payload and success, making it easy to observe patterns.
- **Callback Functions:** Two callback functions are defined: one for the pie chart and another for the scatter plot. These functions take inputs from the dropdown and slider and return the corresponding figure. This ensures real-time updates of the plots based on user interactions, providing an interactive and dynamic dashboard experience.

Predictive Analysis (Classification)

1. Data Preprocessing:

Split the data into features (X) and target labels (Y), standardize X using StandardScaler to ensure all features are on the same scale for model training.

2. Train-Test Split:

Split the data into training (80%) and test (20%) datasets using train_test_split, ensuring reproducibility by setting a random state.

3. Model Selection and Hyperparameter Tuning:

Used GridSearchCV for hyperparameter tuning across different classifiers (Logistic Regression, SVM, Decision Tree, KNN) with cross-validation (CV=10) to optimize each model's performance by selecting the best hyperparameters.

4. Model Evaluation:

Evaluated each model on the test set, calculated accuracy using the .score() method, and visualized performance using confusion matrices to assess classification results.

5. Model Comparison:

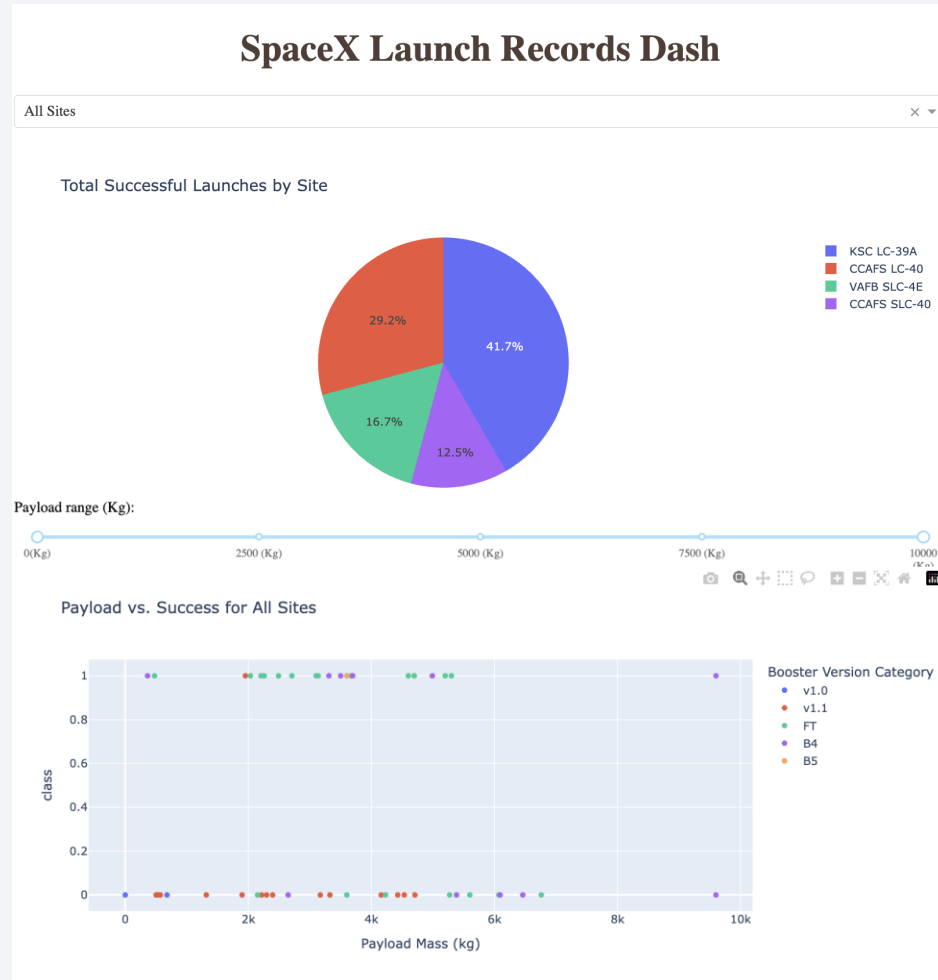
Compared the accuracy of all models (Logistic Regression, SVM, Decision Tree, KNN) on the test set to determine the best performing model.

6. Final Model Selection:

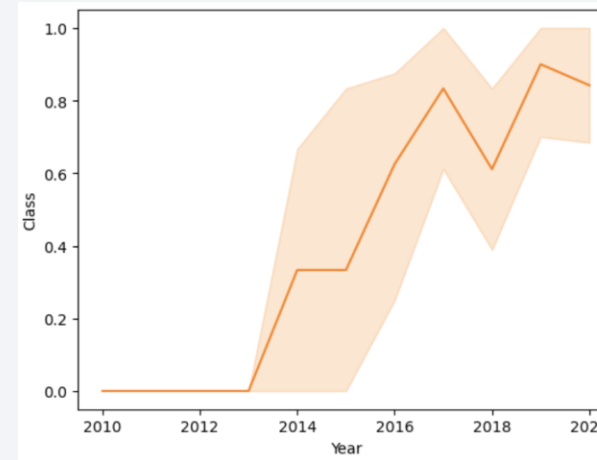
Based on the accuracy scores, selected the best performing model for deployment, and used its best estimator to predict and evaluate on the test set. For this dataset, all the models produced a very similar predictive outcome

Results

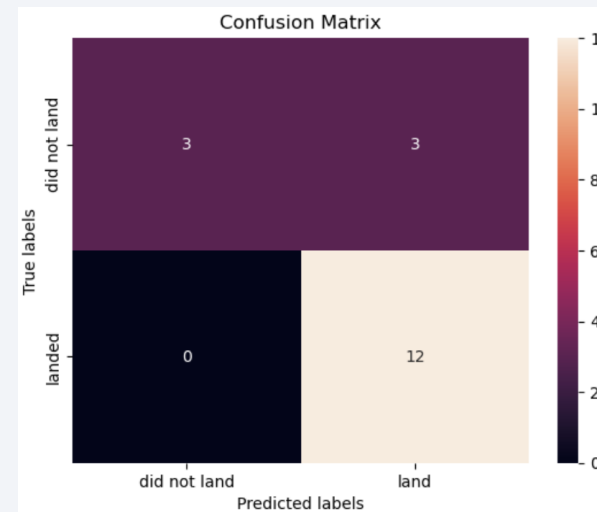
Plotly Interactive Dashboard Screenshot:



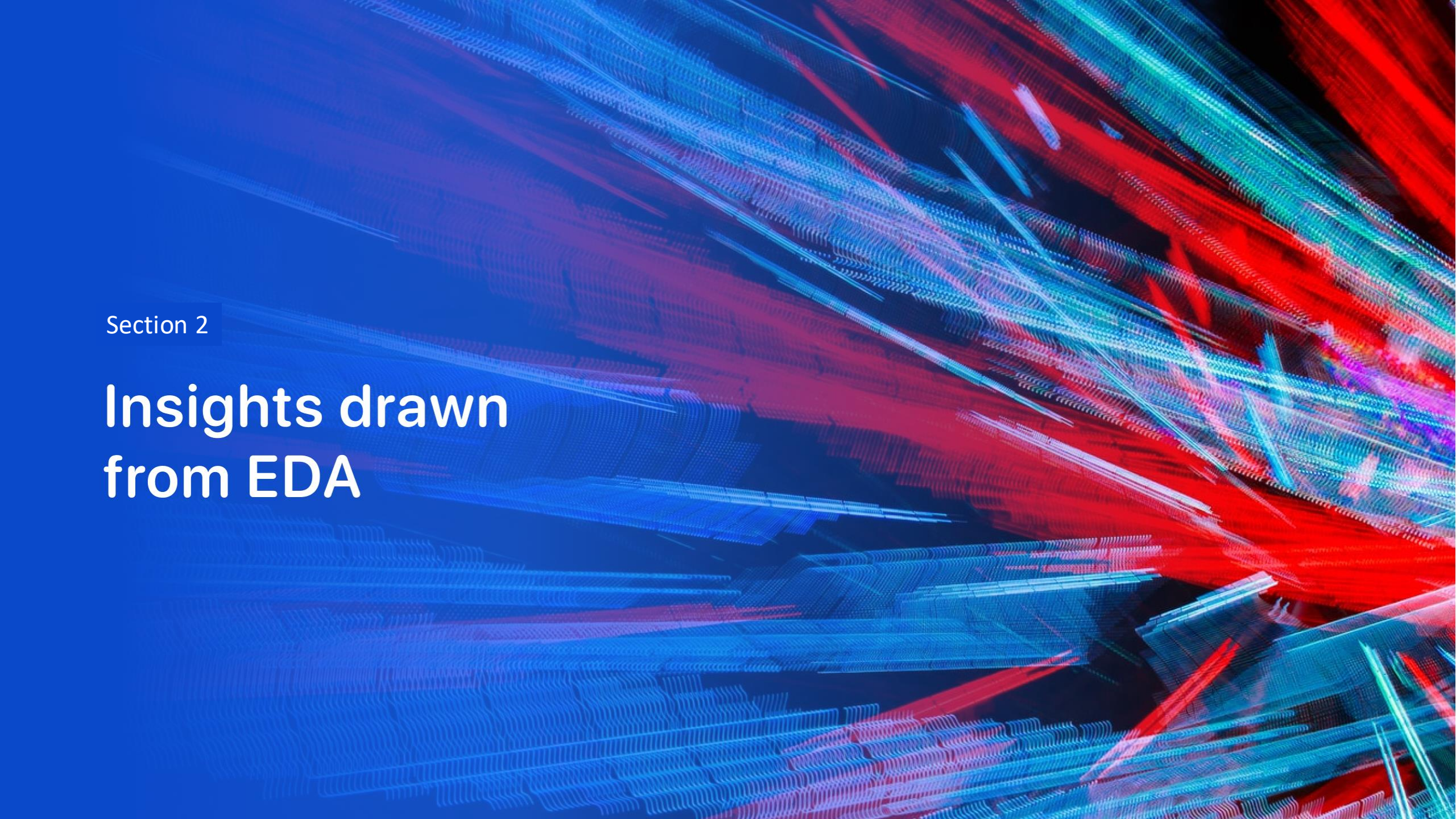
Observations & Findings:



You can observe that the success rate since 2013 kept increasing till 2017 (stable in 2014) and after 2015 it started increasing.



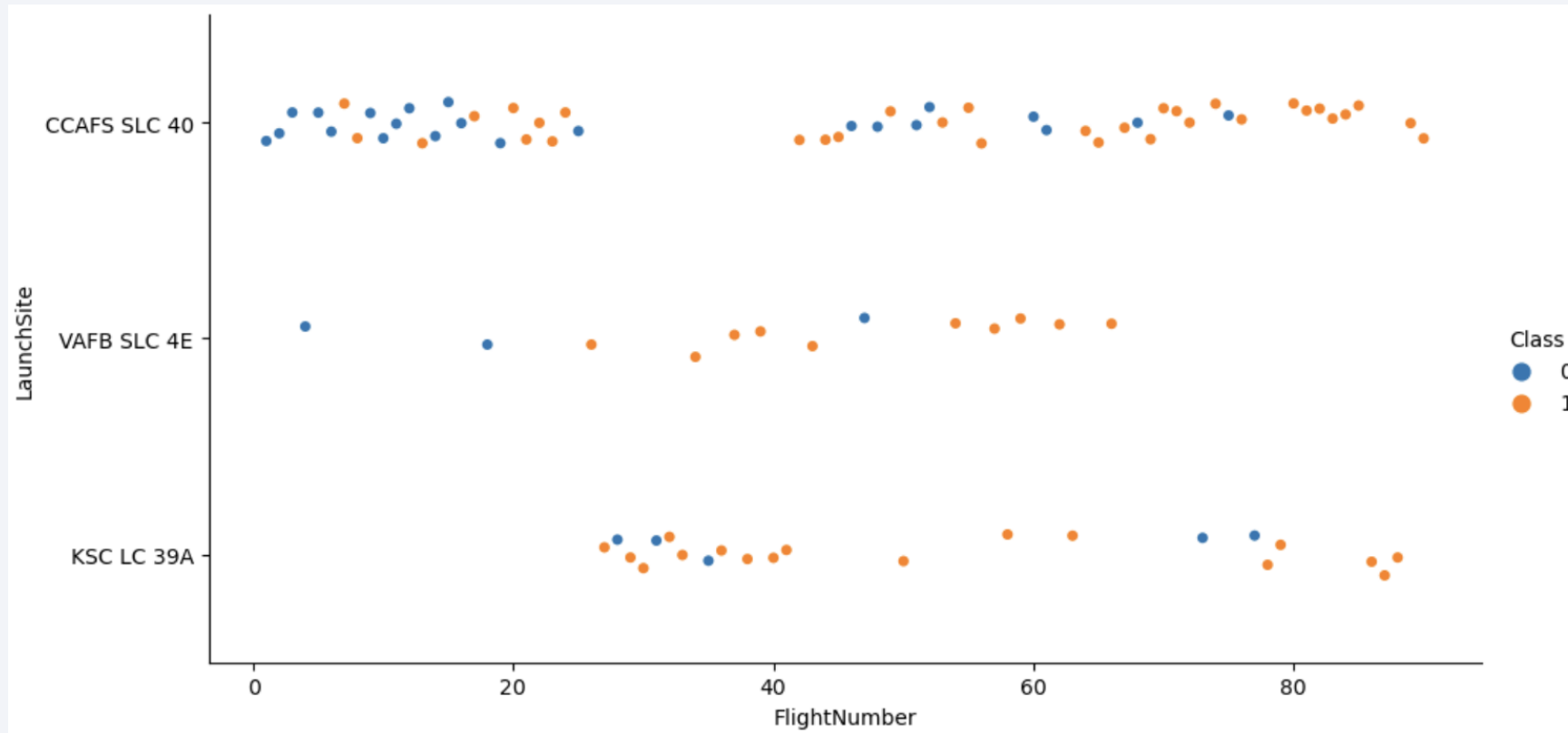
From the 4 Classification models of choice (Logistic Regression, Decision Tree, KNN, SVM) all models yielded the same accuracy at 83.33%

The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

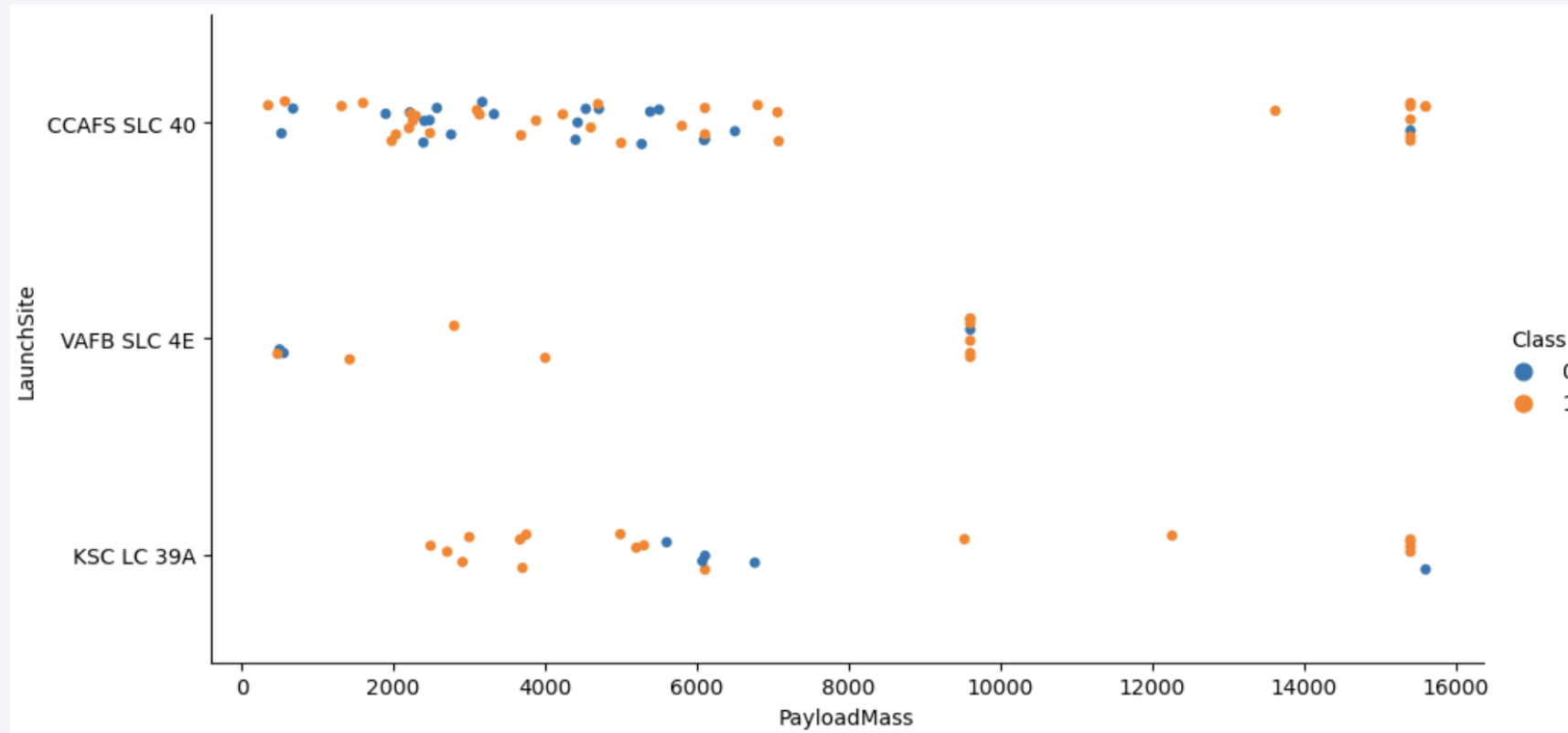
Insights drawn from EDA

Flight Number vs. Launch Site



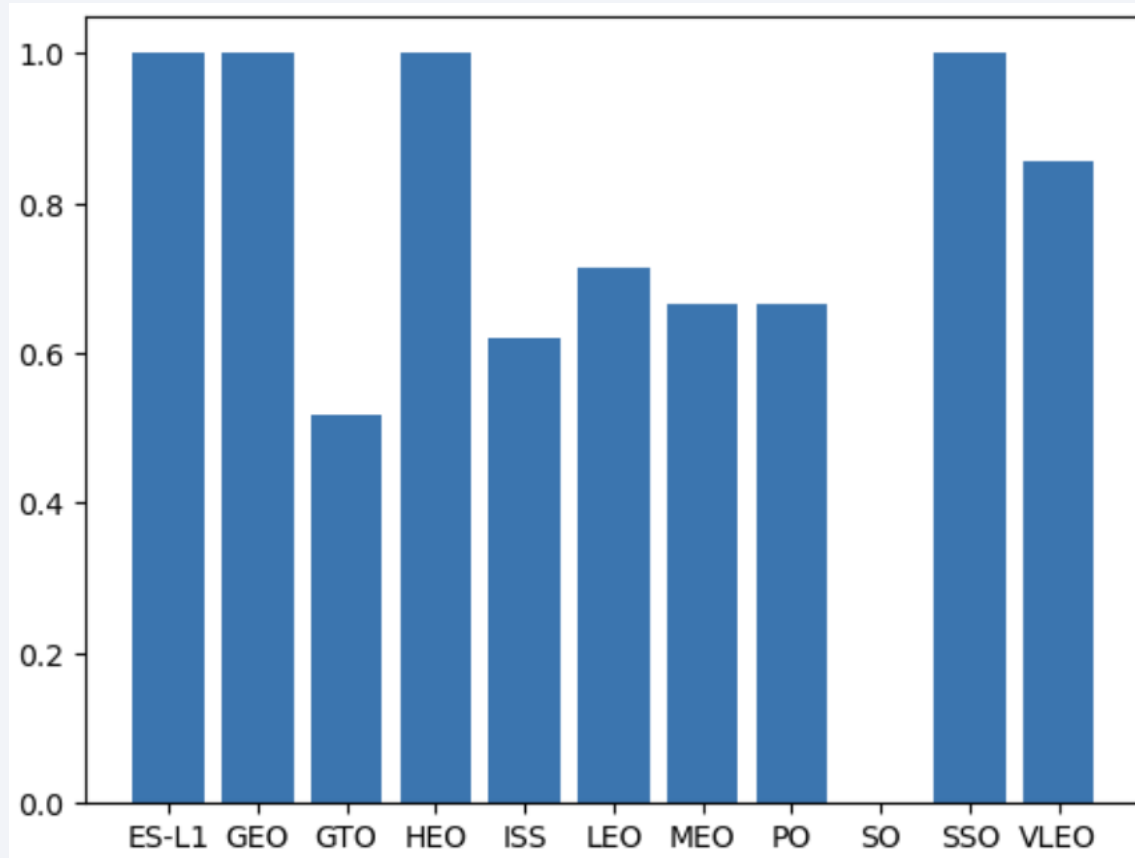
- SLC 40 has the most Falcon 9 rocket launches, albeit with the highest number of failures proportionate to launches
- There seems to be a dramatic improvement in success rate from Flight 25 onwards. Regardless of site

Payload vs. Launch Site



- VAFB SLC 4E has no launches recorded for Payload mass greater the 10k
- There is no obvious correlation between the 2 variables

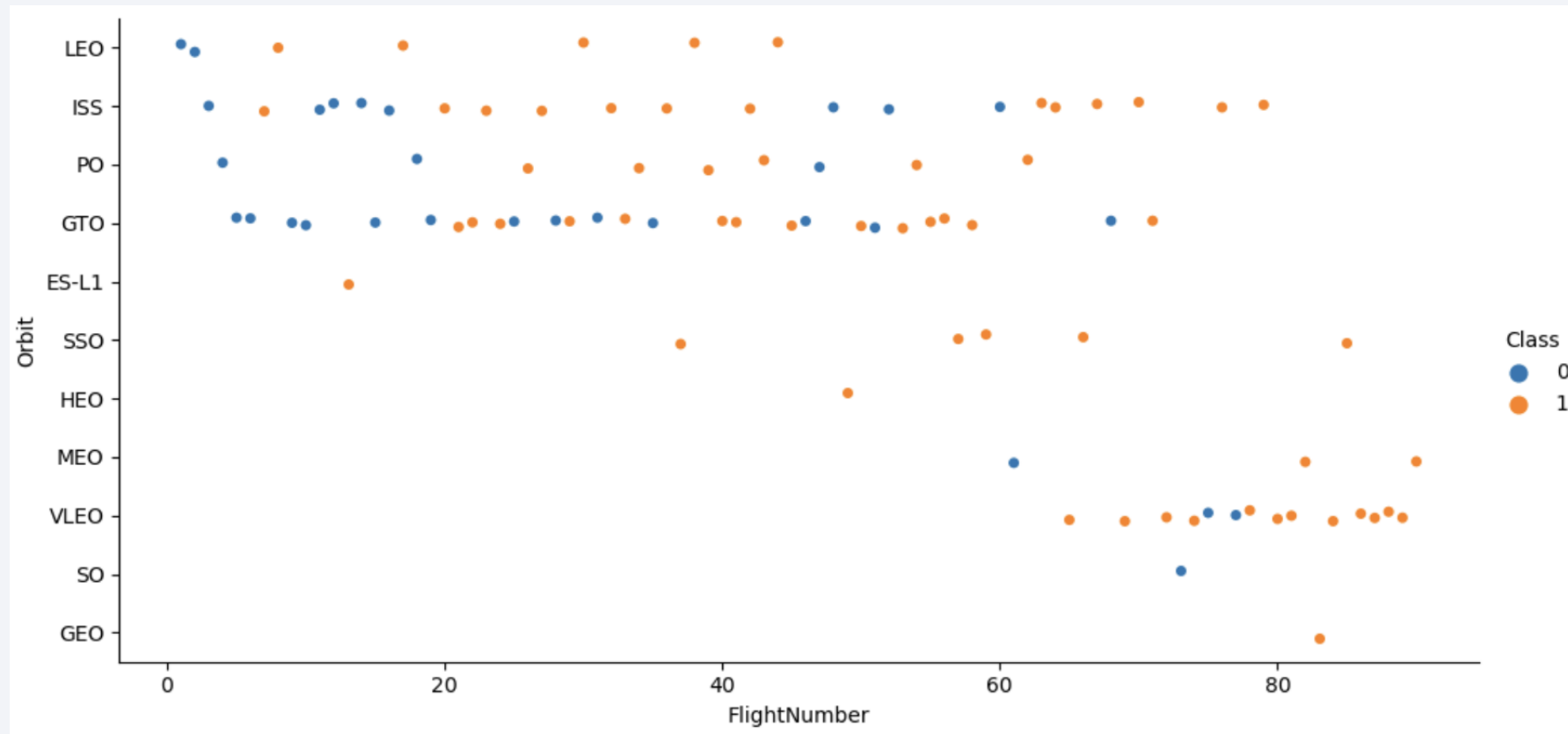
Success Rate vs. Orbit Type



Orbit	Count	Mean
ES-L1	1	1.000000
GEO	1	1.000000
GTO	27	0.518519
HEO	1	1.000000
ISS	21	0.619048
LEO	7	0.714286
MEO	3	0.666667
PO	9	0.666667
SO	1	0.000000
SSO	5	1.000000
VLEO	14	0.857143

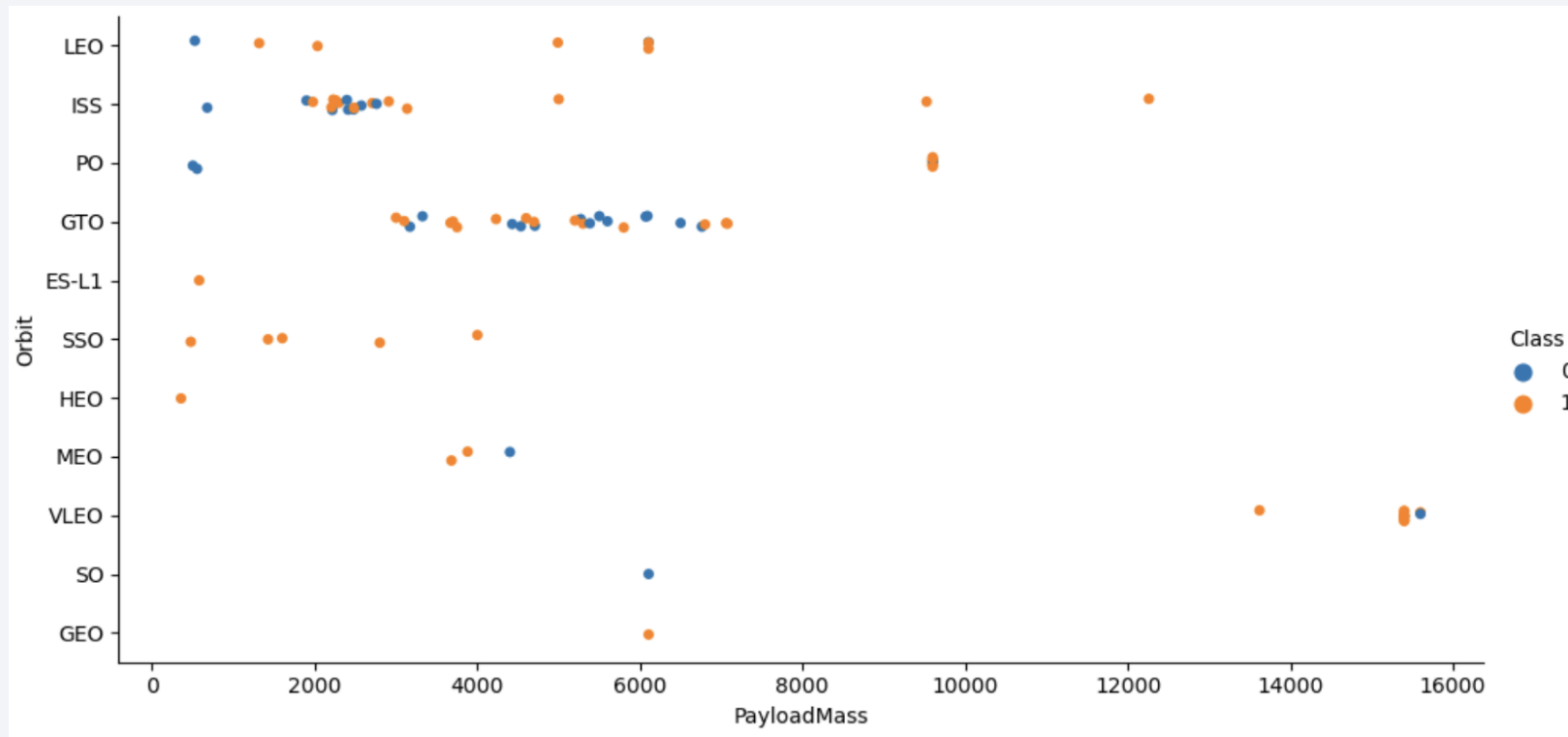
- There are 4 orbits with a 100% success rate, however these contribute to only 8 launches, out of 90 total launches
- There is not enough data to reasonably suggest Orbit types alone contribute to success rate

Flight Number vs. Orbit Type



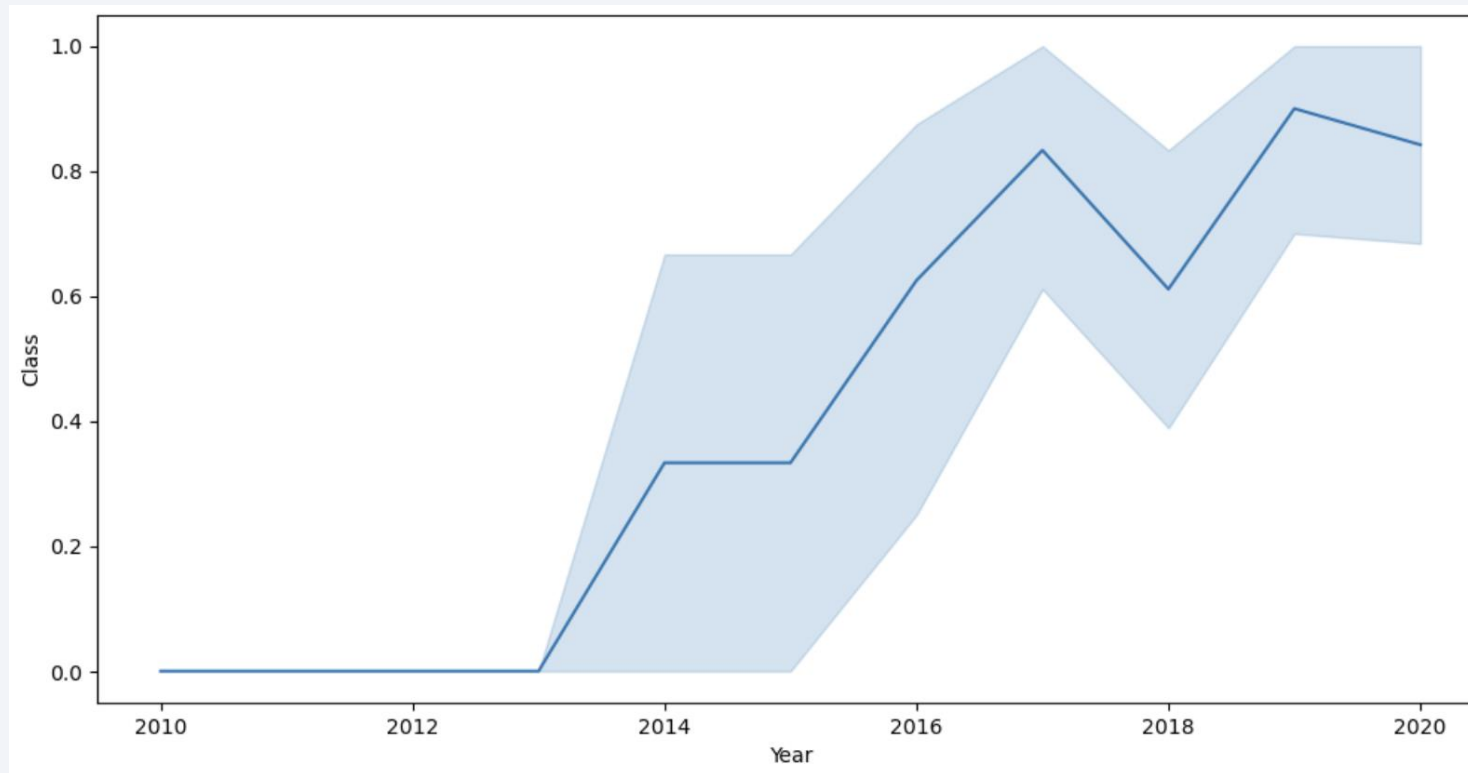
- For LEO, later Flight numbers show increased success rate. Similarly to the LEO, VLEO, SSO but to a lesser extent since there is still noise in the latter orbits
- GTO has no correlation with Flight number in terms of success rate

Payload vs. Orbit Type



- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However for GTO we cannot distinguish this well as both positive landing rate and negative landing(unsuccesful mission) are both there here

Launch Success Yearly Trend



- The data suggests there is an overall increase in success rates on an annual basis, beginning in 2013, bar a few occasions below:
 - 2014 remained flat, noted declines in 2017 and 2019

All Launch Site Names

```
: %sql SELECT DISTINCT "Launch_Site" FROM "SPACEXTABLE";  
* postgresql://chilau:***@localhost:5432/mydb  
  sqlite:///my_data1.db  
4 rows affected.  
:  
  Launch_Site  
KSC LC-39A  
CCAFS LC-40  
CCAFS SLC-40  
VAFB SLC-4E
```

Code Summary:

- Used Sql 'Distinct' clause to find the names of unique launch sites

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM "SPACEXTABLE" WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5
```

```
* postgresql://chilau:***@localhost:5432/mydb  
sqlite:///my_data1.db  
5 rows affected.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Code Summary:

- Used Sql 'Like' operator to find the records of all launch sites beginning 'CCA', and limiting the retrieval to 5 records only.

Total Payload Mass

```
: %sql SELECT SUM("PAYLOAD_MASS__KG_") FROM "SPACEXTABLE" WHERE "Customer" LIKE 'NASA (CRS)'  
* postgresql://chilau:***@localhost:5432/mydb  
  sqlite:///my_data1.db  
1 rows affected.  
:  
sum  
-----  
45596
```

Code Summary:

- Used Sql 'Like' operator to find the records of all launch sites beginning 'CCA', and limiting the retrieval to 5 records only.

Average Payload Mass by F9 v1.1

```
: %sql SELECT AVG("PAYLOAD_MASS__KG_") FROM "SPACEXTABLE" WHERE "Booster_Version" LIKE 'F9 v1.1%'
* postgresql://chilau:***@localhost:5432/mydb
  sqlite:///my_data1.db
1 rows affected.
:
      avg
-----
2534.6666666666666667
```

Code Summary:

- Used Sql 'Like' operator and 'Mean' function to find the average Payload mass (kg) of Booster Versions beginning 'F9 v1.1'.

First Successful Ground Landing Date

```
: %sql SELECT "Date" FROM "SPACEXTABLE" WHERE "Landing_Outcome" LIKE 'Success (ground pad)' ORDER BY "Date" ASC LIMIT(1)
* postgresql://chilau:***@localhost:5432/mydb
  sqlite:///my_data1.db
1 rows affected.
:      Date
-----
2015-12-22
```

Code Summary:

- Used Sql 'Like' operator and 'Order By' and 'Limit' clauses to retrieve the date of the earliest successful ground pad landing.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT * FROM "SPACEXTABLE" WHERE "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000 AND "Landing_Outcome" LIKE 'Success (drone ship)'
```

* postgresql://chilau:***@localhost:5432/mydb
sqlite:///my_data1.db
4 rows affected.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2016-05-06	5:21:00	F9 FT B1022	CCAFS LC-40	JCSAT-14	4696	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2016-08-14	5:26:00	F9 FT B1026	CCAFS LC-40	JCSAT-16	4600	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
2017-10-11	22:53:00	F9 FT B1031.2	KSC LC-39A	SES-11 / EchoStar 105	5200	GTO	SES EchoStar	Success	Success (drone ship)

Code Summary:

- Used Sql 'Like' and 'Between' operators to retrieve the records of successful drone ship landings where the payload mass (kg) is between 4000 and 6000kg inclusive.

Total Number of Successful and Failure Mission Outcomes

```
: %sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") FROM "SPACEXTABLE" GROUP BY "Mission_Outcome"
* postgresql://chilau:***@localhost:5432/mydb
sqlite:///my_data1.db
4 rows affected.
```

Mission_Outcome	count
Success	1
Success (payload status unclear)	1
Success	98
Failure (in flight)	1

Code Summary:

- Used Sql 'Count' operator and 'Group by' clause to summarise the number of Mission Outcomes. Note, Mission Outcome is different from Landing Outcome.

Boosters Carried Maximum Payload

```
%sql SELECT "Booster_Version","PAYLOAD_MASS_KG_", COUNT("Booster_Version") FROM "SPACEXTABLE" \
WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM "SPACEXTABLE") GROUP BY "Booster_Version", "PAYLOAD_MASS_KG_"
```

```
* postgresql://chilau:***@localhost:5432/mydb
sqlite:///my_data1.db
12 rows affected.
```

Booster_Version	PAYLOAD_MASS_KG_	count
F9 B5 B1048.4	15600	1
F9 B5 B1048.5	15600	1
F9 B5 B1049.4	15600	1
F9 B5 B1049.5	15600	1
F9 B5 B1049.7	15600	1
F9 B5 B1051.3	15600	1
F9 B5 B1051.4	15600	1
F9 B5 B1051.6	15600	1
F9 B5 B1056.4	15600	1
F9 B5 B1058.3	15600	1
F9 B5 B1060.2	15600	1
F9 B5 B1060.3	15600	1

Code Summary:

- Used a subquery to find all Booster versions carrying the max payload, and list the Booster version along with payload mass (kg) and count

2015 Launch Records

```
%sql SELECT *, TO_CHAR("Date"::timestamp, 'Month') AS "Month_Name" FROM "SPACEXTABLE" WHERE TO_CHAR("Date"::timestamp, 'YYYY') = '2015' AND "Landing_Outcome" IN ('Failure (drone ship)')
```

```
* postgresql://chilau:***@localhost:5432/mydb  
sqlite:///my_data1.db  
2 rows affected.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome	Month_Name
2015-01-10	9:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)	January
2015-04-14	20:10:00	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)	April

Code Summary:

- Retrieves all columns from the "SPACEXTABLE" for records in 2015 where the "Landing_Outcome" is 'Failure (drone ship)', and adds a new column "Month_Name" representing the month extracted from the "Date" column.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
: %sql SELECT "Landing_Outcome", COUNT("Landing_Outcome") FROM "SPACEXTABLE" WHERE "Landing_Outcome" IN \
(SELECT DISTINCT "Landing_Outcome" FROM "SPACEXTABLE" WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20')\
GROUP BY "Landing_Outcome" ORDER BY "count" DESC
```

```
* postgresql://chilau:***@localhost:5432/mydb
```

```
sqlite:///my_data1.db
```

```
8 rows affected.
```

```
:   Landing_Outcome  count
-----
      No attempt      21
Success (drone ship)  14
Success (ground pad)   9
Failure (drone ship)   5
Controlled (ocean)     5
Uncontrolled (ocean)   2
Failure (parachute)     2
Precluded (drone ship)  1
```

Code Summary:

- Retrieves all columns from the "SPACEXTABLE" for records in 2015 where the "Landing_Outcome" is 'Failure (drone ship)', and adds a new column "Month_Name" representing the month extracted from the "Date" column.

A satellite view of Earth from space, showing the curvature of the planet and the glow of city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Launch Site Locations



Observations:

- All launch sites are close to the coast line
- All launch sites are fairly close to the equator line

Launch Site Clusters & Success Rate

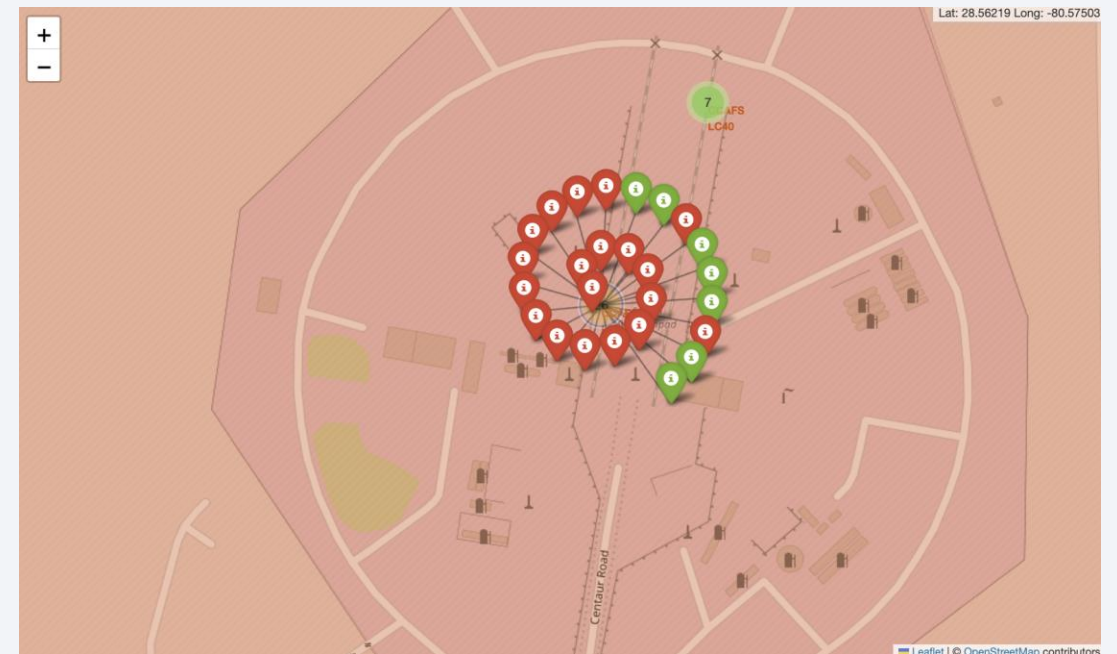
The following shows the launch site clusters on the Folium map, colour coded to indicate launch outcome success.

2 sites are shown with respective launch clusters:

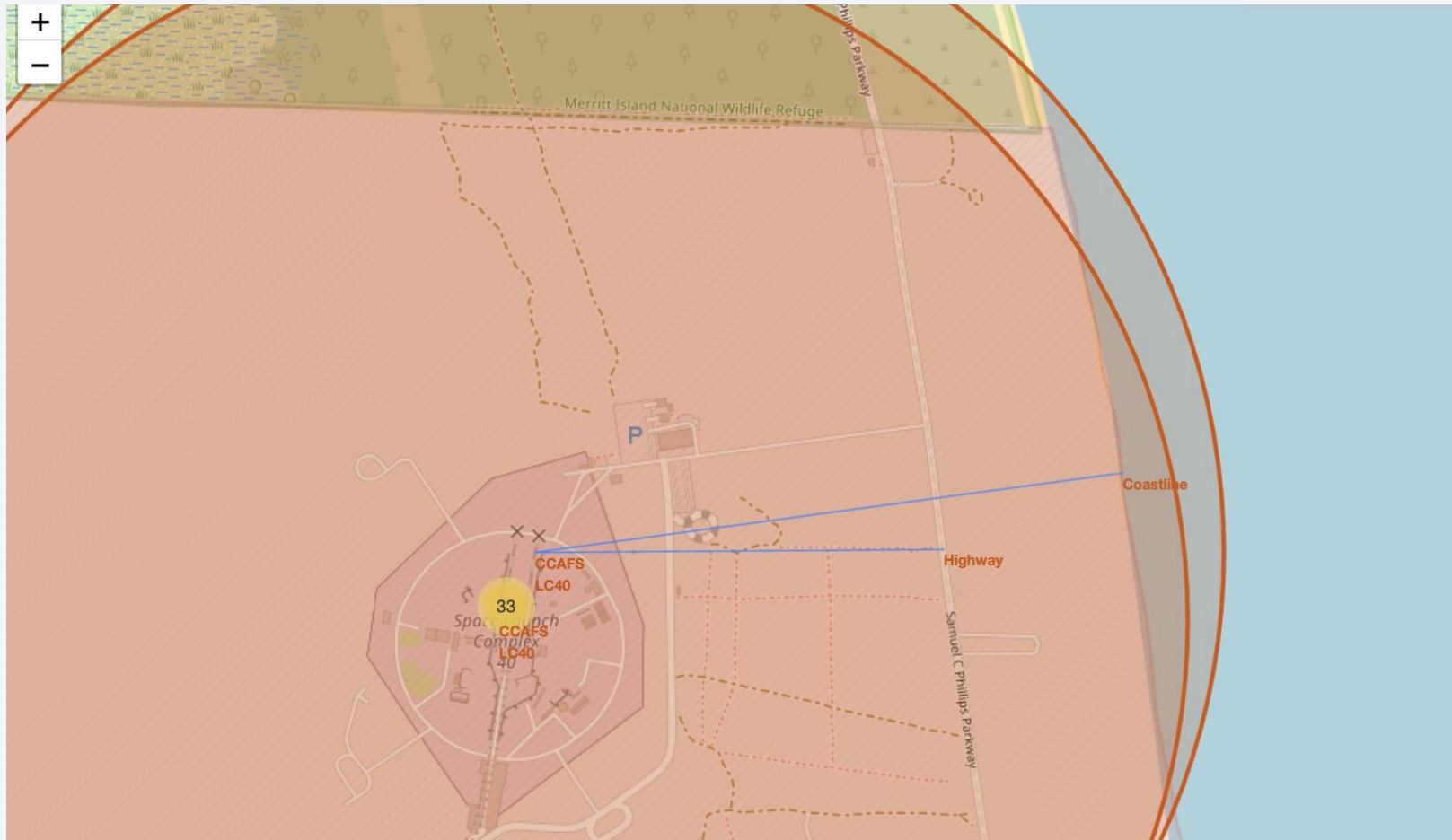
- KSC LC-39A has a success rate of launches at 77%



- CCAFS LC-40 has least success rate of launches at 27%

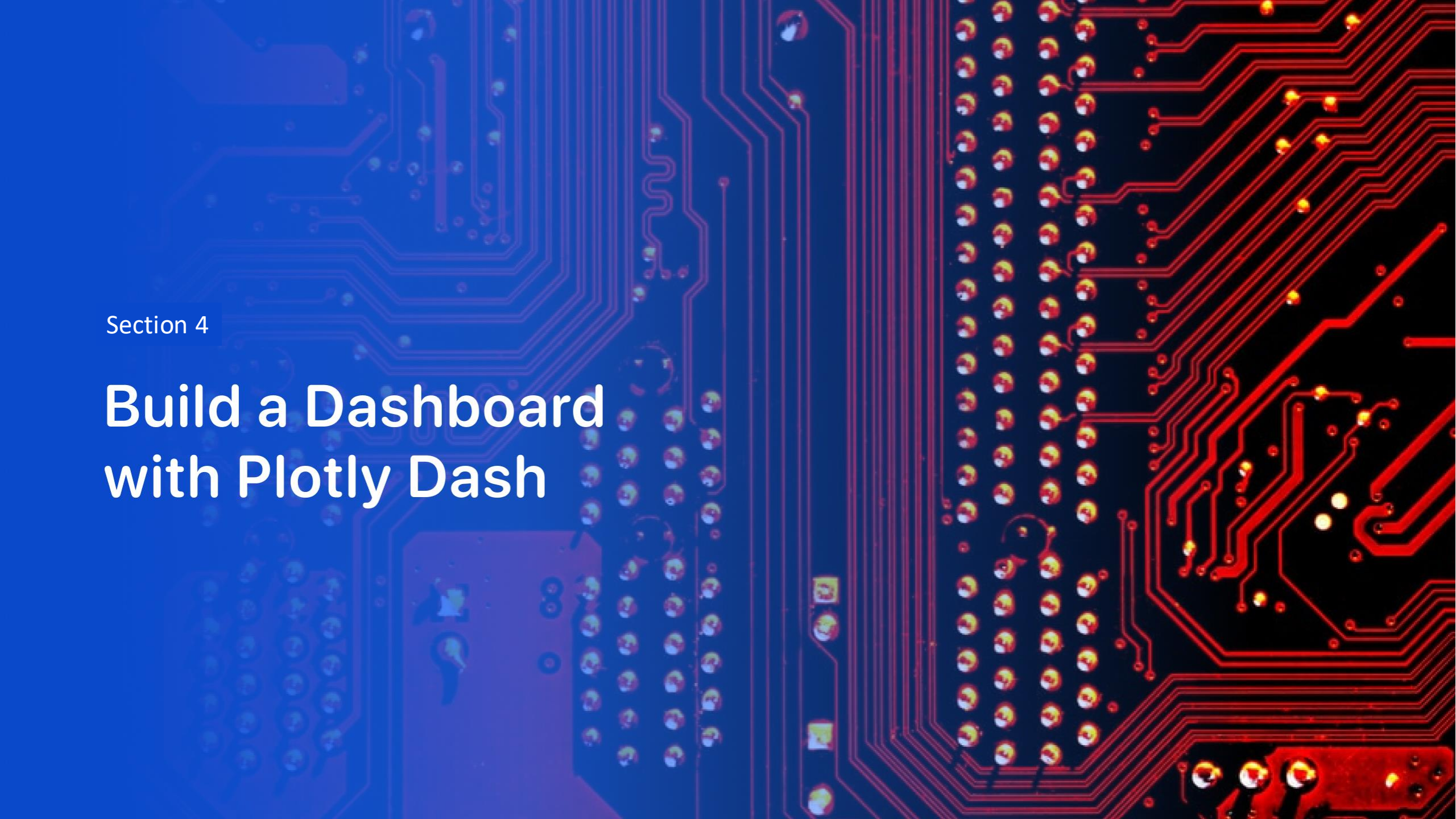


Launch Site Characteristics



All launch sites share common characteristics:

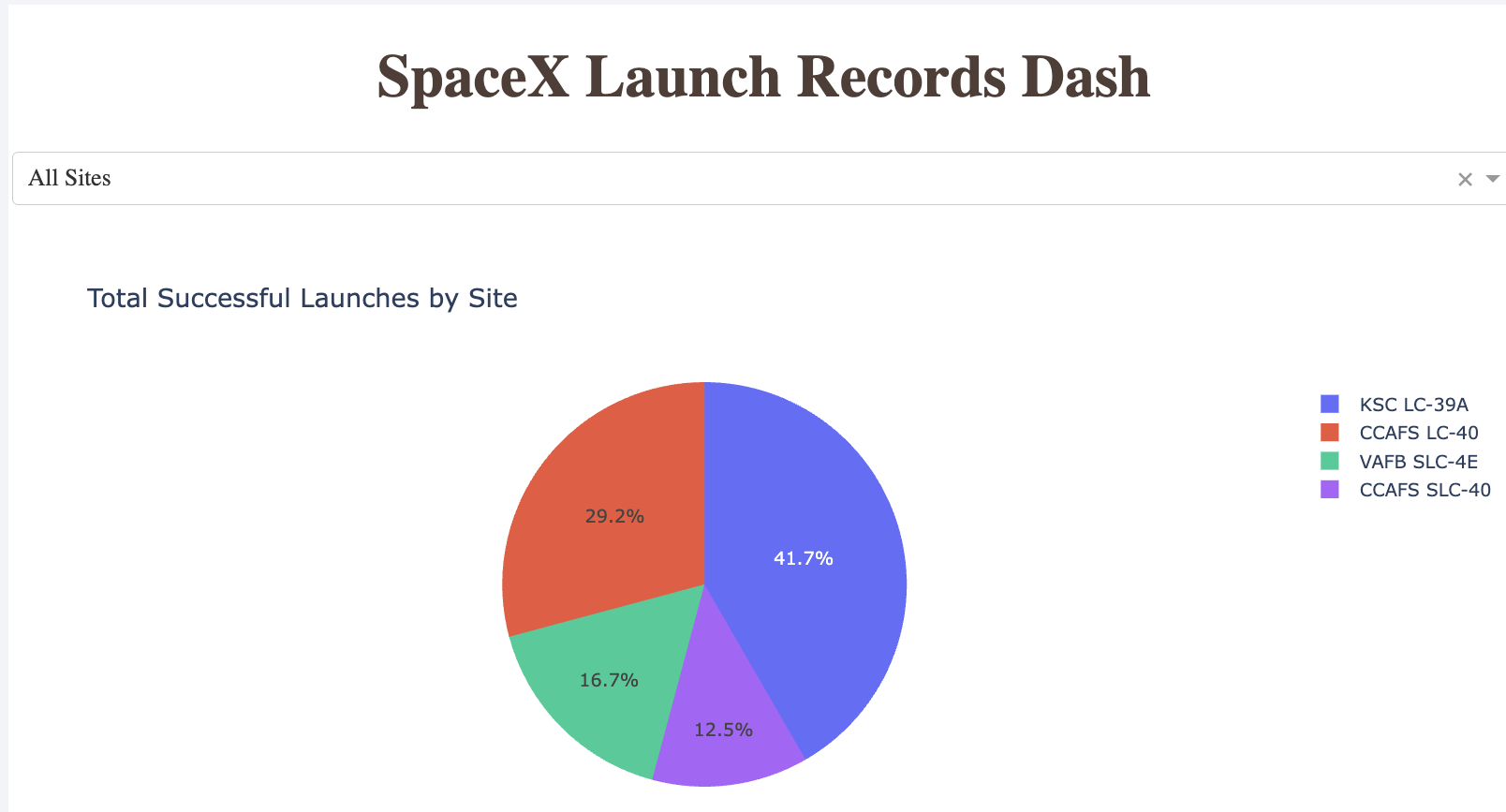
- in close proximity to railways
- in close proximity to highways
- in close proximity to coastline
- launch sites are located far away from cities



Section 4

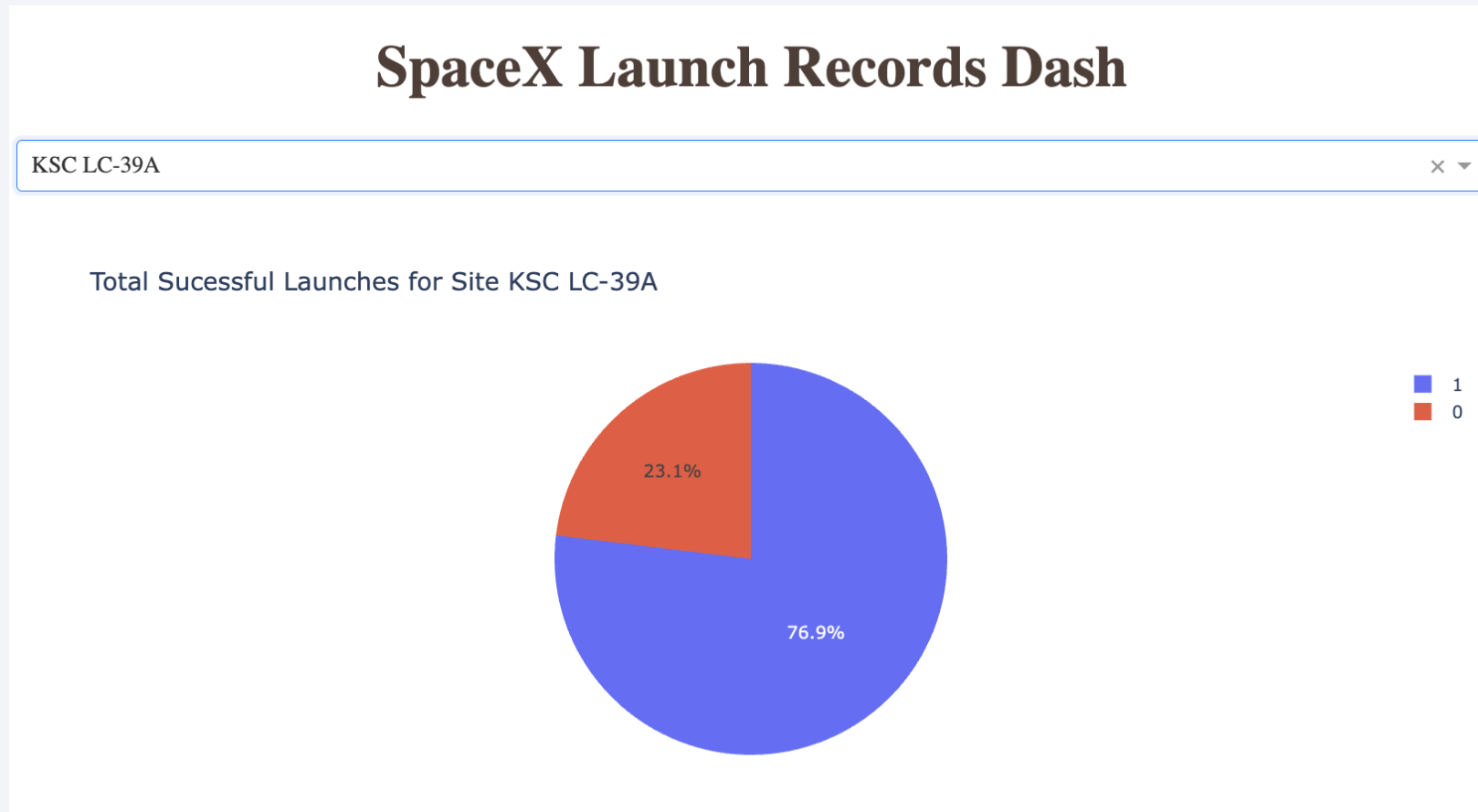
Build a Dashboard with Plotly Dash

Space X Successful Launches



- This section of the dashboard shows the success rate for each launch site as a % of all successful launches.
- Each Site can also be selected to see the success rate at an individual site level

Highest Success Rate Launch Site



- KSC LC-39A has a success rate of launches at 77%
 - Contributing to 42% of all Space X launches within the data set

Booster, Payload Observations



- V1, 1.1 Boosters have the highest failure rate, with a combined successful launch total of 1.
- Payload Mass has no positive impact to both these boosters.



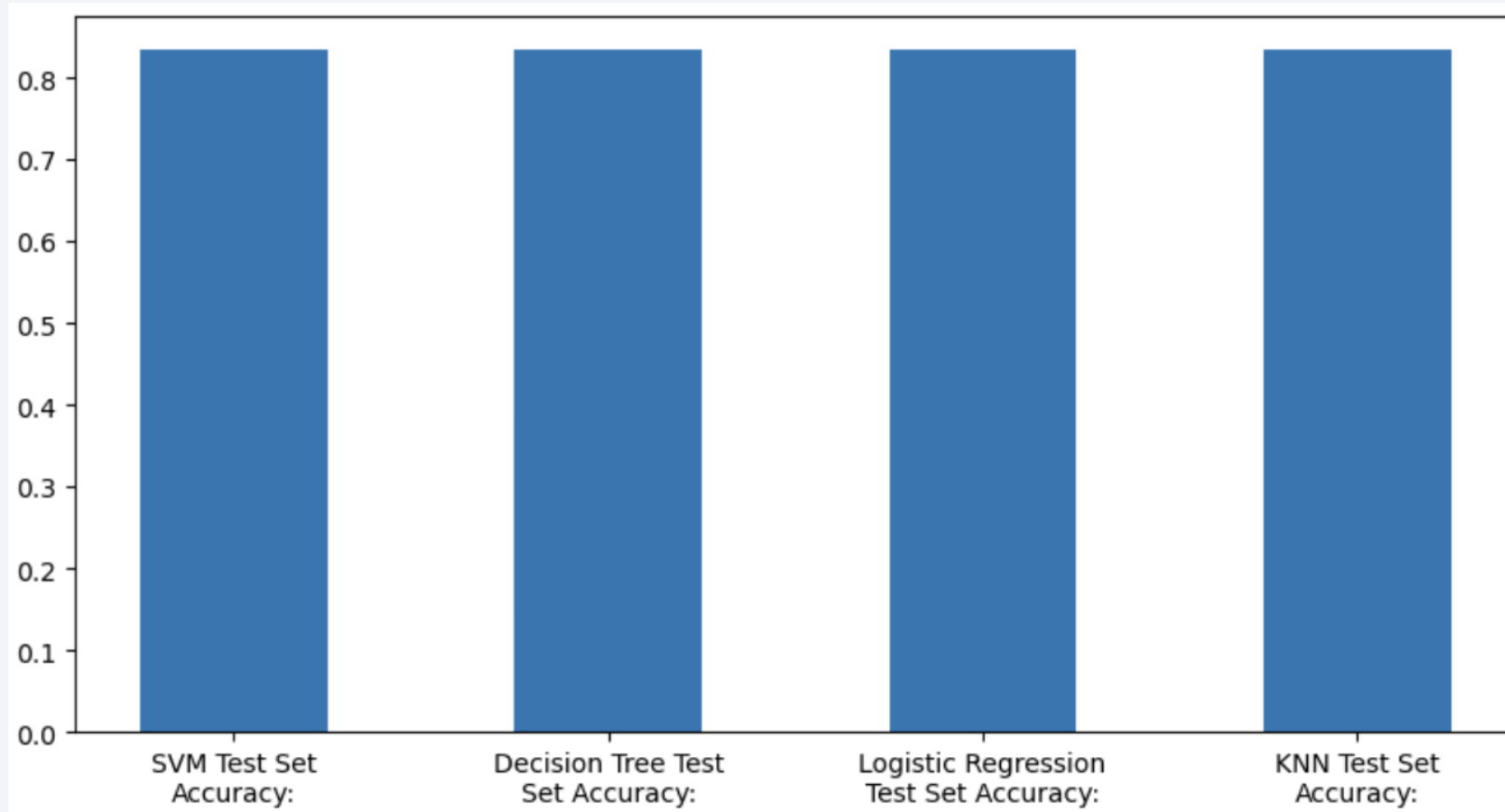
- By filtering out V1-1.1 boosters, we have more suitable data points for the following observations:
 - FT, B4, B9 boosters show more success, predominantly by FT and B4.
 - Payload range (kg) between 2-5K seems to be the ideal range for successful launches



Section 5

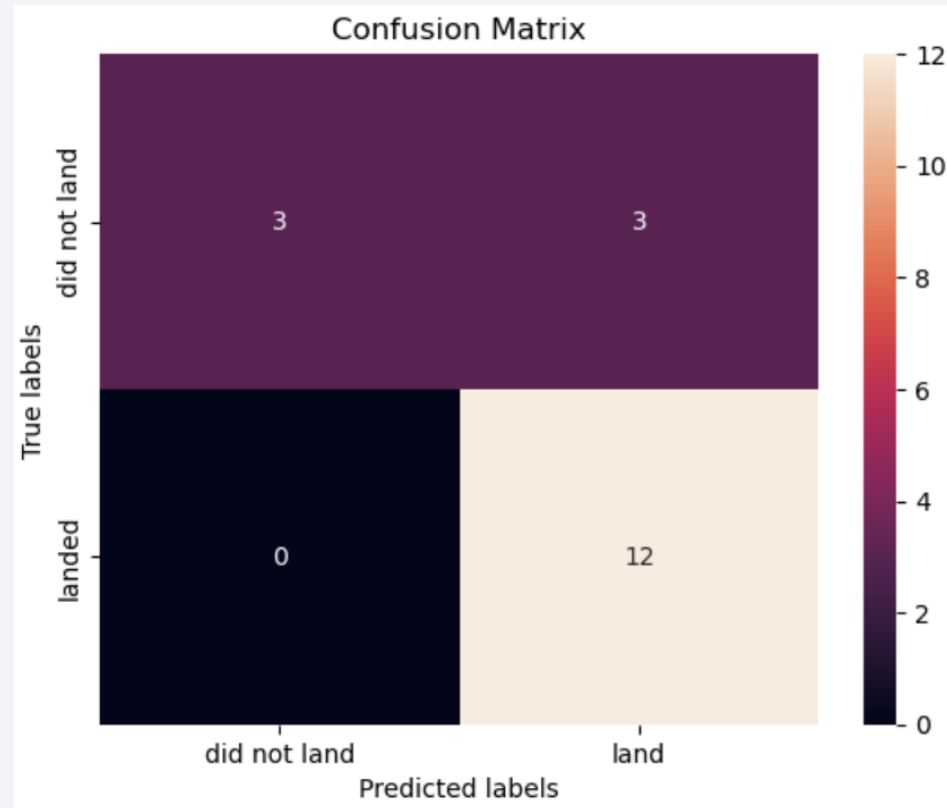
Predictive Analysis (Classification)

Classification Accuracy



- From the 4 Classification models of choice (Logistic Regression, Decision Tree, KNN, SVM) all models yielded the same accuracy at 83.33%

Confusion Matrix



- Since all 4 models yielded the same results, the confusion matrix will remain the same for all models. This matrix is based on the best SVM model via Cross Validation hyperparameter tuning

Conclusions

The Models all exhibited similar results with a strong F1 score more fitted for predicting successful Landings

- Precision: 15 Landed predictions, 12 actually landed
- Recall: of all 12 actual landings, all of them were captured in the predictions.
- The model suggests to be more accurate in predicting landings given an F1 score of 89% (actual 'landed') vs 67% (actual 'did not land')

```
from sklearn.metrics import classification_report
report = classification_report(y_test,y_pred)
print(report)
```

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

Given the data set is mostly Categorical, we can use Permutation Importance to assess the impact of features

- Launch site KSC LC and VAFB SLC data seems to suggest these sites offer increased success rates.
- The most impactful Feature is 'Legs', being 4x more important than launch site KSC LC

Feature	Importance
Legs_True	0.025833333
Legs_False	0.025833333
LaunchSite_KSC LC 39A	0.005694444
LandingPad_5e9e3032383ecb6bb234e7ca	0.005138889
LaunchSite_VAFB SLC 4E	0.004444444
Orbit_VLEO	0.004305556
LandingPad_5e9e3033383ecbb9e534e7cc	0.002916667
Orbit_ISS	0.002777778
Orbit_PO	0.002361111
Serial_B1049	0.001944444
LaunchSite_CCAFS SLC 40	0.001944444
PayloadMass	0.001527778
Serial_B1048	0.00125

Appendix



- By plotting a Catplot of the most important feature 'Legs', the data suggests there is a strong correlation between Legs and a Successful launch (class 1)
- Applying opacity to the markers allows all markers to become visible. It is clearly visible Legs=False are mostly Class 0, whilst Legs=True are mostly Class 1

Thank you!

